Assignment - Math 163

Rotational Inertia of an Annular Cylinder and a Donut-Shaped Solid Body (Torus) using Annular Disk

Md. Shahriar Parvez
Student ID: 1910114

Rotational Inertia

Rotational inertia (or Moment of inertia) is a physical quantity that plays the role of mass of a solid body during its rotational motion. It is constant for a particular solid body and a particular rotation axis. Mathematically, it can be represented as:

$$I = \sum_{i=1}^{n} m_i r_i^2 \tag{1}$$

Where:

- m_i = Mass of a particle of that solid body.
- r_i = Perpendicular distance of that particle from the rotation axis.

If the solid body consists of a great many adjacent particles, then we can write Equation (1) in the integral form as:

$$I = \int r^2 \, dm \tag{2}$$

Problem Statement

1. Calculate the rotational inertia of an annular cylinder about its central axis.
2. Using that result, calculate the rotational inertia of a donut-shaped rigid body (solid torus) about its central axis.

Solution

(i) Rotational Inertia of an Annular Cylinder

Let:

- M = Mass of the annular cylinder
- R = Outer radius of the annular cylinder
- r = Inner radius of the annular cylinder
- h = Height of the cylinder

Consider a cylindrical shell element of thickness dx situated at a radial distance x from the rotation axis. The rotational inertia of this differential segment is:

$$dI = x^2 \, dM$$

Where dM is the mass of the segment. Using uniform mass density:

$$dM = 2\pi x h \, dx \times \frac{M}{\pi(R^2 - r^2)h} = \frac{2M}{R^2 - r^2} x \, dx$$

Substituting dM into dI:

$$dI = \frac{2M}{R^2 - r^2} x^3 \, dx$$

Integrating from $x = r$ to $x = R$:

$$\int_0^I dI = \int_r^R \frac{2M}{R^2 - r^2} x^3 \, dx$$

$$I = \frac{2M}{R^2 - r^2} \left[\frac{x^4}{4} \right]_r^R = \frac{2M}{R^2 - r^2} \frac{R^4 - r^4}{4}$$

$$I = \frac{1}{2} M (R^2 + r^2)$$

This is the desired rotational inertia of the annular cylinder about its central axis.

(ii) Rotational Inertia of a Donut-Shaped Body (Solid Torus)

Let:

- R = Distance from the central axis to the center of the cross-section
- r = Radius of the circular cross-section
- M = Mass of the rigid body

For a horizontal slice at a distance x from the cross-sectional center with thickness dx:

$$\begin{aligned} a &= \sqrt{r^2 - x^2} \\ R_2 &= R + a = R + \sqrt{r^2 - x^2} \quad \text{(Outer radius)} \\ R_1 &= R - a = R - \sqrt{r^2 - x^2} \quad \text{(Inner radius)} \end{aligned}$$

1. Volume Calculation

The differential volume dV of a slice of thickness dx is:

$$dV = \pi (R_2^2 - R_1^2) \, dx = \pi \left[(R + \sqrt{r^2 - x^2})^2 - (R - \sqrt{r^2 - x^2})^2 \right] dx$$

$$dV = 4\pi R \sqrt{r^2 - x^2} \, dx$$

Integrating over $x \in [-r, r]$:

$$V = \int_{-r}^{r} 4\pi R \sqrt{r^2 - x^2} \, dx = 8\pi R \int_0^r \sqrt{r^2 - x^2} \, dx$$

Since $\int_0^r \sqrt{r^2 - x^2} \, dx = \frac{\pi r^2}{4}$:

$$V = 8\pi R \times \frac{\pi r^2}{4} = 2\pi^2 R r^2$$

2. Rotational Inertia Integration

Using the formula for the rotational inertia of an annular cylinder from part (i), the differential rotational inertia dI of the slice is:

$$dI = \frac{1}{2} dM (R_2^2 + R_1^2)$$

Where $dM = dV \times \frac{M}{V}$ and $(R_2^2 + R_1^2) = 2(R^2 + r^2 - x^2)$:

$$dI = \frac{1}{2} \left(4\pi R \sqrt{r^2 - x^2} \, dx \right) \left(\frac{M}{2\pi^2 R r^2} \right) \cdot 2(R^2 + r^2 - x^2)$$

$$dI = \frac{2M}{\pi r^2} (R^2 + r^2 - x^2) \sqrt{r^2 - x^2} \, dx$$

Let $b^2 = R^2 + r^2$:

$$dI = \frac{2M}{\pi r^2}\left(b^2\sqrt{r^2-x^2} - x^2\sqrt{r^2-x^2}\right)dx$$

Integrating from $x = -r$ to $x = r$:

$$I = \frac{2M}{\pi r^2}\left(\int_{-r}^{r} b^2\sqrt{r^2-x^2}\,dx - \int_{-r}^{r} x^2\sqrt{r^2-x^2}\,dx\right)$$

Exploiting symmetry for even functions:

$$I = \frac{4M}{\pi r^2}\left(b^2\int_{0}^{r} \sqrt{r^2-x^2}\,dx - \int_{0}^{r} x^2\sqrt{r^2-x^2}\,dx\right)$$

$$I = \frac{4M}{\pi r^2}\left(b^2\cdot\frac{\pi r^2}{4} - I_0\right) = M(R^2+r^2) - \frac{4M}{\pi r^2}I_0 \quad \text{— (3)}$$

3. Evaluating I_0 using Gamma Function

Let $I_0 = \int_0^r x^2\sqrt{r^2-x^2}\,dx$. Substituting $x = r\sin\theta \implies dx = r\cos\theta\,d\theta$:

$$I_0 = \int_0^{\frac{\pi}{2}} (r^2\sin^2\theta)(r\cos\theta)(r\cos\theta)\,d\theta = r^4\int_0^{\frac{\pi}{2}} \sin^2\theta\cos^2\theta\,d\theta$$

Using the Beta/Gamma Function identity $\int_0^{\pi/2}\sin^m\theta\cos^n\theta\,d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)}$:

$$I_0 = r^4\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{3}{2}\right)}{2\Gamma(3)} = r^4\frac{\left(\frac{1}{2}\sqrt{\pi}\right)^2}{2\times 2!} = \frac{\pi r^4}{16}$$

Substituting I_0 back into Equation (3):

$$I = M(R^2+r^2) - \frac{4M}{\pi r^2}\left(\frac{\pi r^4}{16}\right)$$

$$I = MR^2 + Mr^2 - \frac{1}{4}Mr^2$$

$$\mathbf{I = MR^2 + \frac{3}{4}Mr^2 = M\left(R^2 + \frac{3}{4}r^2\right)}$$

Which is the desired rotational inertia of the donut-shaped rigid body.